EPFL

NII Inter-University Research Institute Corporation Research Organization of Information and Systems National Institute of Informatics

MASTER PROJECT · UNDERSTANDING THE LABELS

Triple barriers & meta-labeling

Where the ground truth comes from,
how the trade is evaluated, and what M2 adds.

AUTHOR Elie Bruno · NII Tokyo

READING López de Prado · Advances in Financial Machine Learning · Chapter 3

UPDATE 29 October 2026 · UP M1 Sharpe −2.5744 · 48 / 48 fits reviewed

Which dates belong to which results?

Configured cutoffs · daily horizon = 7 bars · all dates below use YYYY-MM-DD.

Configuration	Data start	Train ends	Validation ends	Data ends
Earlier trial · separate PDF	2024-07-01	2025-05-30	2025-10-01	2026-01-25
Pablo GPU8 · now copied	2024-07-01	2025-08-31	2025-12-31	2026-08-01

Actual sample dates · after context and split rules

New daily inputs · Pablo's configuration
Train: 2024-06-30 – 2025-08-31
Val-Cal: 2025-09-08 – 2025-10-18
Val-Opt: 2025-10-26 – 2025-12-31
Test: 2026-01-08 – 2026-07-18

Other evidence and historical context
Label diagnostics: 2024-07-01 – 2025-08-24
Historical AG / Kumo Test · separate PDF:
2025-10-09 – 2026-01-18
Separate LightGBM Test:
2026-01-01 – 2026-09-23

The original loader creates a first training window on **2024-06-30**, one day before the configured start. Its context and embargo conventions are preserved. The realised Test dates match Pablo's saved reports.

GPU8 config observed 2026-10-27; daily input checks 2026-10-28; saved earlier trial, diagnostics and benchmark dates. Full LightGBM train/validation ranges accompany its result slides.

VERIFIED ON 28 OCTOBER 2026 · M1 BASELINE

The reproduced UP baseline has Sharpe −2.5744

Test: 2026-01-08 – 2026-07-18 · spread equity · 7-day horizon · fee 0.2%.

M1 side	Return	Sharpe	Max drawdown	Trades
UP	−54.25%	**−2.5744**	−57.99%	1045
DOWN	30.44%	1.7423	−17.04%	2580

12 / 12
Target / direction baseline checks passed

27
Weekly return observations from 192 daily bars

−39.86%
Buy-and-hold return on the same Test period

Buy-and-hold also lost money in this period. All six targets share M1's Test trades. **Four-backend comparison: 48 / 48 fits reviewed.** AutoGluon, Kumo, TabPFN and Random Forest are all complete.

Independent daily data preparation reproduces saved GPU8 M1 metrics within 0.0001; existing UP multigranularity cache also checked. Spread equity smooths trade returns; Sharpe uses ×√(365.25/7). The inspected Test is exploratory.

Return and drawdown explain the “−45%” figure

Same 1045 UP trades · 2026-01-08 – 2026-07-18 · different equity calculations.

Equity calculation	Total return	Max drawdown	Sharpe
Spread · original baseline	−54.25%	−57.99%	−2.57
Cash · open positions at cost	−43.60%	−45.35%	−2.79
Cash · current market prices	−43.66%	−48.75%	−2.14

Saved cash report

−43.60% is the endpoint loss.
−45.35% is the worst fall from a prior peak.
Open positions remain at entry value.

Current cash code

Open positions follow current closing prices.
Interim losses become visible: **−48.75% drawdown**.
Use the same valuation rule for comparisons.

Replay of Pablo’s saved UP trade CSV on 2026-10-28; cash exposure 0.10 in both rows. Saved cash metrics match positions held at cost. Final-day entries can remain open: cash figures are endpoint equity, including the current marking fee.

Pablo’s saved UP M2 Sharpes are also negative

Test entries: 2026-01-08 – 2026-07-18 · 1d · TP labels · spread-equity engine.

Strategy	UP return	UP Sharpe	DOWN return	DOWN Sharpe
M1 · all calls	−54.25%	**−2.57**	30.44%	1.74
AutoGluon	−54.22%	**−2.36**	22.36%	1.58
CTTS	−58.65%	**−2.23**	23.36%	1.45
TabPFN	−21.50%	**−1.07**	20.70%	1.70
TabICL	−57.91%	**−2.43**	35.95%	1.91

These are **Pablo’s saved TP results**. The completed 48-fit matrix uses **fresh TP controls for each backend**. M1, dates and the backtest match; the new M2 scores differ from these saved classifiers.

Saved GPU8 outputs fetched 2026-10-27; 48 replayed metrics agree within 0.0001. New models are separate fits on Izar, with pinned principal packages; hardware, Python patch release and fitted ensembles can differ. Exact M2 reproduction is not established.

01 · GROUND TRUTH

What is the ground truth based on?

The observed future price path, evaluated under a rule fixed at entry.

AT ENTRY · FIX THE QUESTION

M1 proposes a trade
Example: UP → buy
Entry price: 100

→

Specify the trade
Profit at 104 · stop at 97
Time limit: day 7

→

Freeze the rule
Fix the exit policy
and cost assumptions.

AFTERWARDS · OBSERVE THE ANSWER

Actual market prices
100 → 101 → 97 → …
Stop hit on day 2.

→

Calculate the outcome
Buy at 100, exit at 97
−3% − 0.2% costs = −3.2%

→

Ground truth: z = 0
This proposed trade lost money.
M2 will learn to predict z.

M1 supplies the proposed direction. Actual future prices supply the outcome.
The ground-truth label is calculated from that outcome.

AFML §§3.2–3.6. Illustrative long trade; assumed fill at the barrier and total cost of 0.2%.

Two methods answer two different questions

DEFINE THE OUTCOME

When does the trade end?

Triple-barrier method

Follow the price until profit-taking, stop-loss or the time limit.

Output: an **exit time** and a **realized return**.

PREDICT THE OUTCOME

Should we take M1's trade?

Meta-labeling

Learn whether a proposed long or short trade will have a successful outcome.

Output: a **probability** used to accept, skip or size the trade.

They can be combined. Meta-labeling also works with a fixed exit at day 7.

AFML §3.4 (barriers), §§3.6–3.7 (meta-labeling).

The day-7 price can hide an earlier loss



Hold until day 7

Buy at 100 → sell at 103
Net return: **+2.8%**
Meta-label: **1**

Exit at the stop

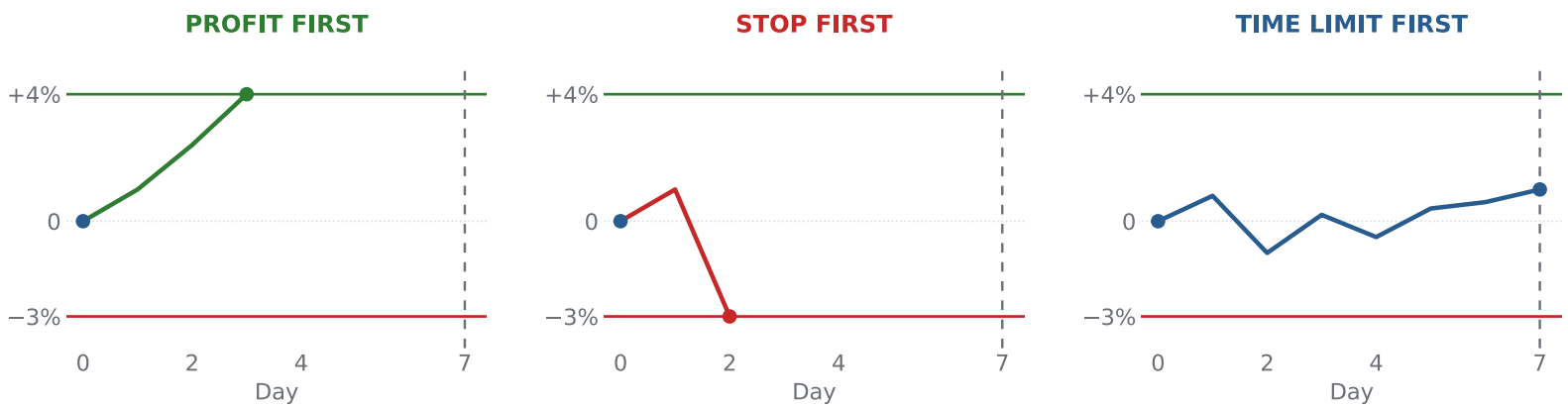
Buy at 100 → sell at 97
Net return: **−3.2%**
Meta-label: **0**

Same M1 forecast, same market path. A different exit rule gives a different ground truth.

Illustration, not market data. Profit at 104; stop at 97; time limit day 7; cost 0.2%.

Close the trade at whichever barrier comes first

Example: buy at 100 · take profit at 104 · stop at 97 · close by day 7.



$$\tau = \min\left(t_{\text{profit}}, t_{\text{stop}}, t_{\text{expiry}}\right)$$

The first touch determines the exit. The return at that exit determines the label.

AFML §3.4, pp. 45–47. Three illustrative paths, shown as gross returns on a long trade.

Set the distances using information available at entry

Scale the distances by volatility

Let v_0 be a return-volatility estimate known at entry.

Profit target: $+av_0$
Stop-loss: $-bv_0$
Time limit: h bars

a and b may differ.

Our experiment's choice

$v_0 = \sigma_0\sqrt{h}, \quad h = 7$

σ_0: EWMA estimate of daily-return volatility, using past prices.

$\sqrt{h}$ is a horizon-scaling approximation. The method does not require it.

For a **short** trade, profit is below the entry price and the stop is above it.
Apply the barriers to the trade's return, with its direction accounted for.

AFML §3.3 and Snippets 3.1–3.2. Our symmetric experiments use a = b = k; the +4%/−3% examples are illustrative.

A direction label and a meta-label mean different things

Suppose price goes from **100 to 99** by expiry, with no earlier barrier touch.

LABEL THE PRICE MOVEMENT

What did the market do?

Direction label: DOWN

The price fell by 1%.

In a directional target, this can be encoded as -1.

LABEL THE PROPOSED TRADE

Did M1's proposed trade pay?

M1 side	Net return	z
Long	−1.2%	**0**
Short	+0.8%	**1**

A falling price can be a **successful short**.

Meta-label **0 means "this proposal failed"**. It does not mean "take the opposite side".

AFML §3.5 versus §3.6, especially Snippets 3.5 and 3.7. Net returns use the illustrative 0.2% cost.

Meta-label 1 means a positive net trade return

$$g = s\left(\frac{P_\tau}{P_0} - 1\right) - c \qquad z = \begin{cases} 1 \text{ if } \ g > 0 \\ 0 \text{ otherwise} \end{cases}$$

s: +1 long / −1 short · P_0: entry · P_τ: exit · c: total cost

Long trade starting at 100	Exit price	Net return	z
Profit reached first	104	+3.8%	**1**
Stop reached first	97	−3.2%	**0**
Expiry, small gain	101	+0.8%	**1**
Expiry, loss	99	−1.2%	**0**
Expiry, gain below costs	100.1	−0.1%	**0**

> Expiry is an exit condition. It can produce either meta-label.

AFML Snippet 3.7 labels positive side-adjusted returns as 1. Subtracting c = 0.2% is our cost-aware extension.

Build the labels first; then train M2 to predict them

1 · Save the inputs

At entry: market features
+ M1's forecast and side.

→

2 · Observe the future

Follow actual prices until
profit, stop or time limit.

→

3 · Calculate z

1 if net return > 0;
0 otherwise.

↓ Repeat for historical M1 proposals

Training examples

Entry-time inputs → eventual target z
Both profitable and unprofitable proposals.

→

Fit M2

Learn which entry-time conditions
are associated with z = 1.

Future prices are used to calculate the **training target**.
The **inputs** are the features and M1 outputs that were available at entry.

AFML §§3.6–3.7. Labels can be calculated for hypothetical trades; actual execution is not required.

At decision time, the outcome is still unknown

A new M1 proposal
Today's features + forecast
M1 chooses long or short.

→

Trained M2
Estimates p = P(z = 1)
for that proposed trade.

→

Decision rule
Take M1's side or skip.
Optional: determine size.

Example · take
M1: SHORT · M2: p = 0.75
With a 0.60 cutoff: take the SHORT.

Example · skip
M1: SHORT · M2: p = 0.40
With the same cutoff: no position.

M2 predicts the chance of success; it does not receive the future price path or the label.
The 0.60 cutoff is illustrative. Choose the decision rule on validation data.

AFML §§3.6–3.7; probability-based bet sizing in §10.3. Calibration and payoff sizes matter for using p.

M2 can filter the weaker proposals from M1

M1 PROPOSES

100 proposed trades

55 profitable · **45** unprofitable

55%
Win rate if all proposals are taken

M2 SELECTS

40 accepted trades

32 profitable · **8** unprofitable

80%
Win rate among accepted proposals

The 60 skipped proposals include **23 missed winners** and **37 avoided losers**.

The aim is better selection, with fewer trades. This improvement must be measured.
Win rate alone does not determine profit: the sizes of wins and losses also matter.

Illustrative counts, not experimental results. AFML §3.7 motivates the precision–recall trade-off.

We already use meta-labeling. Barriers change the target.

FIXED-HORIZON META-LABEL

Our current dataset

M1: Kronos proposes UP or DOWN.
M2: predicts whether that proposal succeeds.

Exit: **day 7**

$$z_{\text{fixed}} = \mathbb{1}\left[s\left(\frac{P_7}{P_0} - 1\right) > c\right]$$

“Would holding this side for 7 days pay?”

TRIPLE-BARRIER META-LABEL

With triple-barrier labels

M1 and M2 keep the same roles.
The outcome follows a different trade.

Exit: **first profit, stop or expiry**

$$z_{\text{TB}} = \mathbb{1}\left[s\left(\frac{P_\tau}{P_0} - 1\right) > c\right]$$

“Would this side pay under these exits?”

The added value is a target that reflects the chosen exit policy.
When M2 is meant to predict trade profitability, label that same trade.

Current dataset: Synthetic-Generation/data/complete/README.md. Barrier labels: analysis/label_counts.py.

A larger day-7 threshold asks a different question

Target	When evaluated	What makes z = 1?
Current meta-label	Day 7	The proposed side earns more than costs.
Larger return hurdle	Day 7	The proposed side earns more than a chosen volatility-scaled threshold.
Triple-barrier meta-label	First exit	The proposed side earns more than costs at that exit.

A volatility threshold changes **how large the gain must be**.
Triple barriers change **when the trade ends**. Both change the definition of success.

AFML §§3.2–3.4. Our existing threshold experiment uses $s \cdot r_7 > k\sigma_o\sqrt{7}$; it is a separate target.

The existing experiment has not established a gain

LABEL COMPARISON

5.9% of labels change

With symmetric $\pm\sigma_0\sqrt{7}$ barriers:

Positive labels: **54.7% → 52.7%**

254,952 historical daily Kronos calls.

LABEL-ONLY BENCHMARK

Δ Sharpe: −0.06

Barrier-trained M2 versus the current target, with both exiting at day 7.

95% interval: **−0.57 to +0.30**

This interval includes zero.

These results concern this setup. They do not establish that barriers never help.
A changed definition of success needs evaluation under the intended trading policy.

Separate LightGBM benchmark, 3 seeds; 20% validation cutoff. Entry dates: train 2024-07-01 – 2025-08-24 validation 2025-09-01 – 2025-12-24 Test 2026-01-01 – 2026-09-23. Saved label_counts.csv and label_benchmark.csv.

Use the same label rule on real and synthetic paths

Real example
Observed past + actual future

Synthetic example
Generated past + generated future

→

Same labeling procedure
Run frozen M1 on the past.
Evaluate its side on the future.
Use the same exits and costs.

→

Train M2
Inputs at entry
+ calculated z

A synthetic label is correct **for the generated path**.
Whether that helps on real markets must be checked on held-out real data.

The generated future is used for labels; M1 and M2 inputs use the generated past only.

Application to this thesis. The labeling rule follows AFML ch. 3; improvement from augmentation is a research question.

Observed prices → trade outcome → meta-label → M2

Triple barriers
Specify when the proposed trade ends.

→

Ground-truth label
Did that trade earn a positive net return?

→

M2
Predict the chance of that outcome for a new proposal.

What is the addition?

Triple barriers add the **price path and exit policy** to the target.
Meta-labeling adds **selection, and potentially sizing**, to M1's direction.

> First define the trade we want to run. Then calculate its labels and test whether M2 helps select it.

AFML §§3.4–3.7. Triple barriers are a labeling method; improved prediction and profit require evidence.

READING GUIDE

Where these ideas appear in the book

Question	López de Prado (2018), AFML
Why does the price path matter?	§§3.2–3.4 · pp. 43–47
What is a directional label?	§3.5 · pp. 48–50 · Snippet 3.5
What is a meta-label?	§3.6 · pp. 50–51 · Snippet 3.7
What does the second model add?	§3.7 · pp. 51–53
How can probability inform size?	§10.3 · pp. 142–144
How do we avoid overlapping-label leakage?	§7.4 · pp. 105–109

Page numbers refer to the printed book. All schematic price paths and filtering counts are illustrative. The 0.2% cost and 7-day experiments are our conventions, not universal requirements of the method.

A

Appendix: experiments and implementation

Supporting results and assumptions for discussion.

Tighter barriers change more training labels

Target	Positive	Negative	Labels changed
Day 7	139,453	115,499	—
Barrier k = 0.5	128,936	126,016	23.5%
Barrier k = 1	134,295	120,657	5.9%
Barrier k = 1.5	137,022	117,930	2.0%
Barrier k = 2	138,265	116,687	0.9%

A changed label is a changed target, not evidence of a better predictor.

254,952 calls; all periods pooled. Entry dates: train 2024-07-01 – 2025-08-24 validation 2025-09-01 – 2025-12-24 Test 2026-01-01 – 2026-09-23. High/low touches; same-bar ties = stop. Saved label_counts.csv.

Changing M2's target while still exiting at day 7

M2 target	Test trades	Test Sharpe	Δ Sharpe · 95% interval
Day 7	20,976	−0.73	Reference
Barrier k = 0.5	17,386	−1.16	−0.44 · [−1.28, +0.25]
Barrier k = 1	20,398	−0.79	−0.06 · [−0.57, +0.30]
Barrier k = 1.5	20,904	−0.74	−0.01 · [−0.30, +0.19]
Barrier k = 2	20,896	−0.72	+0.00 · [−0.12, +0.10]

Each cutoff selects the top 20% on validation and is then held fixed.
The number of accepted test trades can differ. None of these intervals excludes zero.

Test entries: 2026-01-01 – 2026-09-23. Saved label_benchmark.csv; LightGBM per side; predictions averaged over 3 seeds; paired 10-day block bootstrap.

Changing the exit is a separate experiment

Comparison on the test period	Δ Sharpe	95% interval
M1 only: volatility barriers vs day 7	−0.67	−1.17 to −0.25
M1 only: volatility barriers vs fixed ±10%	+0.49	+0.04 to +1.06
M2: barrier label + exit vs current label + exit	−0.23	−1.32 to +0.55

Changing labels and changing execution answer different experimental questions.
These saved comparisons do not establish an improvement from the full proposal.

Test entries: 2026-01-01 – 2026-09-23. Saved benchmark_ci.json; 10-day block bootstrap. M2 rows use a single-seed model, unlike the label-only table.

A valid label also needs realistic timing and execution

Information and validation

- Features and volatility use information available at entry.
- Generate M1 calls out of sample if M1 is trained locally.
- Purge labels that overlap held-out periods; embargo where needed.
- Select rules on validation; use fresh data for confirmation.

Price observation and fills

- The book's snippets check observed closes.
- Our experiments check daily highs and lows.
- Both barriers in one candle: order is unknown; our code assumes stop first.
- Barrier fills and fixed 0.2% costs are simplifying assumptions.

AFML Snippet 3.2 and §7.4; local tb.py and benchmark.py. Gaps, slippage, funding and borrow costs need appropriate execution modeling.

The binary target is not a position size

From prediction to position

M2 produces a score interpreted as $p = P(z = 1 \mid \text{inputs}, s)$.

A separate rule maps that score to a nonnegative amount $q(p)$.

$$\text{position} = s \cdot q(p), \quad q(p) \geq 0$$

Zero size = skip; M1 keeps the side.

Probability alone is insufficient

With net gain G on a win and net loss magnitude L on a loss:

$$E[g] = pG - (1-p)L$$

$$E[g] > 0 \Leftrightarrow p > \frac{L}{G+L}$$

Toy example: gain 1, lose 2 → break-even probability = **66.7%**.

Use calibrated probabilities, realistic payoffs and a risk budget.
A 50% probability cutoff is not automatically the trading break-even point.

AFML §3.6 and §10.3. Expected-value equation is an illustrative two-payoff calculation, not the book's sizing formula.

READING THE COMPARISON

TP means true-positive meta-labeling

Original TP asks whether a Kronos proposal made money at day 7, **after the 0.2% fee**.

Label 1 · true positive

The proposed UP or DOWN trade was profitable after the fee at day 7.

Label 0 · unsuccessful proposal

Its signed day-7 return was zero or negative after the fee.

Column	M2 training target	Purpose
TP	Original seven-day true-positive label	Original target baseline for each backend
TB ref	Same TP label; TB eligibility filter	Control with matched training eligibility
TB k	Profit / stop / time-barrier outcome	Test the effect of a different target

TP is not a take-profit order. M2 predicts success from the available features; it does not know the future label. All reported backtests still execute at day 7, including models trained on TB targets.

Original conversion: pred=1 and lab=1 → meta_label=1; pred=1 and lab=0 → 0. The signed day-7 rule agrees with all 18,230 complete-horizon proposals in the 40 daily CSVs. RF = Random Forest; TabPFN is the original GPU8 backend.

Four backends, six targets: 48 / 48 fits reviewed

Test: 2026-01-08 – 2026-07-18 · TP = true positive · fixed day-7 execution.
M1 Sharpe: **UP −2.5744** · DOWN 1.7423. RF = Random Forest.

M2 fit	TP	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−2.199	−2.375*	−2.246	−2.376*	−2.305	−2.280
AutoGluon · DOWN	0.339	1.452	1.822	1.357	1.383	1.080
Kumo · UP	−1.983	−2.255	−2.463	−2.714	−2.038	−2.225
Kumo · DOWN	1.884	2.001	−0.278	1.731	1.849	1.730
TabPFN · UP	−2.527	−2.570	−2.353*	−2.594	−2.445	−2.519
TabPFN · DOWN	0.882	0.988	1.569	1.046	0.829	0.797
RF · UP	−1.800	−1.517	−2.222	−1.779	−1.388	−1.810
RF · DOWN	1.201	1.246	−0.473	−0.256	1.222	0.825

All UP scores remain negative. RF at k = 1.5 has the least-negative UP Sharpe, **−1.388**. The highest observed DOWN score is **Kumo TB ref: 2.001**, retaining TP labels.

48 fits; 288 return/drawdown/Sharpe metrics replayed within 0.0001. Annualized spread equity, 27 horizon returns. * = utility constraint failed, fallback threshold 0.5. TabPFN/RF use frozen GPU8 tuning parameters. Exploratory Test.

How much does Sharpe change versus the fresh TP fit?

Δ Sharpe = target Sharpe − TP Sharpe · Test 2026-01-08 – 2026-07-18.

M2 fit	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−0.176	−0.047	−0.178	−0.106	−0.081
AutoGluon · DOWN	+1.114	+1.483	+1.019	+1.044	+0.742
Kumo · UP	−0.272	−0.480	−0.731	−0.055	−0.242
Kumo · DOWN	+0.117	−2.162	−0.153	−0.035	−0.154
TabPFN · UP	−0.043	+0.174	−0.067	+0.082	+0.008
TabPFN · DOWN	+0.106	+0.687	+0.164	−0.054	−0.085
RF · UP	+0.283	−0.422	+0.021	+0.412	−0.010
RF · DOWN	+0.045	−1.675	−1.458	+0.020	−0.377

At k = 0.5, DOWN gains are **AutoGluon +1.483** and **TabPFN +0.687** versus TP. RF UP at k = 1.5 improves by **+0.412**, but its Sharpe remains negative.

This contrast includes the eligibility change: TB ref and TB-k remove **13 UP / 64 DOWN** early TON training proposals. The matched comparison follows.

Same realized Test trades, fees, features and fixed day-7 execution. Fresh TP fits differ from Pablo's saved M2 fits. Deltas are descriptive; this inspected Test does not select an optimal k.

TB label effects depend on backend and direction

Δ Sharpe = TB-k Sharpe − TB-ref Sharpe · TB ref keeps TP labels on the TB-eligible samples.

M2 fit	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	+0.128	−0.002	+0.069	+0.094
AutoGluon · DOWN	+0.370	−0.095	−0.069	−0.372
Kumo · UP	−0.208	−0.459	+0.217	+0.030
Kumo · DOWN	−2.279	−0.270	−0.152	−0.271
TabPFN · UP	+0.217	−0.024	+0.125	+0.051
TabPFN · DOWN	+0.580	+0.058	−0.160	−0.191
RF · UP	−0.704	−0.261	+0.130	−0.293
RF · DOWN	−1.719	−1.502	−0.024	−0.421

TabPFN · DOWN · k = 0.5

Sharpe **1.569**: **+0.580** versus TB ref.
Return **+34.78%**; max drawdown **−17.51%**.
Still below M1's DOWN Sharpe of 1.742.

Random Forest · UP · k = 1.5

Sharpe **−1.388**: **+0.130** versus TB ref.
Return **−34.61%**: improvement still loses money.
RF DOWN k = 1.5 trails TB ref by 0.024.

All TB variants share training eligibility and Test trades. This is a label comparison with day-7 execution, not early barrier-exit trading. One fit per cell; 27 horizon returns; no significance or optimal-k claim.